

# MediaPipe based Gesture Recognition System for English Letters

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## ABSTRACT

Gesture recognition is widely used in communication, virtual reality, smart cockpit, smart home, etc. This paper proposes an application system based on MediaPipe for intelligent recognition of 26 letter gestures in English. The system uses images captured by a monocular camera, reads images using OpenCV, creates a hand detection model, uses MediaPipe to identify 21 key points of the hand, constructs a gesture recognition module for English letters, and recognizes letters through rough gesture recognition algorithm and fine gesture recognition algorithm. The experimental results show that the recognition accuracy of all English letters in this system is more than 81%, and the recall rate is more than 83%, which can be applied to gesture recognition related applications of English letters.

## CCS CONCEPTS

• **Software and its engineering**; • **Software creation and management**; • **Software development techniques**; • **Object oriented development**;

## KEYWORDS

MediaPipe, OpenCV, Gesture recognition, English letter recognition

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## 1 INTRODUCTION

Gesture recognition is an important form of human-computer interaction, which has broad application prospects in communication with hearing impaired people, virtual reality, smart cockpit, smart home, etc. Gestures are mainly composed of hand actions. They express different meanings by drawing different trajectories and actions in the air. As a part of human social understanding and expression, gesture recognition is different from mouse keyboard

interaction. With the help of computer vision and artificial intelligence technology, gesture detection is easier for people to use computers and networks.

As early as the 1980s, people began to study gesture recognition. According to the form of data, it can be divided into image, video and depth image, and these three kinds of data have achieved high recognition rate in the field of gesture recognition. At present, there are mainly three kinds of gesture data acquisition devices: data gloves, somatosensory devices, and digital cameras.

Data glove is the earliest collection device used for gesture recognition [1]. There are many sensors on the data glove. Through analysis and research on various data obtained by different sensors, corresponding feature parameters are extracted and combined to identify hand shape, direction, movement track and other information. And the collected signal can achieve high accuracy without any pre-processing, which is convenient for training. And it has a good speed of reflection. Therefore, data gloves have great advantages in the research of gesture recognition systems with large vocabulary. Using data gloves to recognize gestures, many scholars have done a lot of research [2–9], and also made some achievements. The earliest gesture recognition is based on data gloves, and the research on data gloves has been decades old. The data glove has a good performance in gesture recognition accuracy and real-time performance by virtue of the feature that it can directly obtain data with high accuracy. However, the cost of data gloves is too high, it is inconvenient to wear, and it does not conform to people's usage habits, so it is difficult to promote and apply them in practice. At present, they are only used in laboratories.

In 2010, Microsoft released Kinect somatosensory device. Kinect is a device that can recognize 25 key points of the human body, which makes the gesture recognition field have new input devices and enriches the research methods of gesture recognition. Kinect can obtain the depth image information, and can quickly and accurately obtain the key points of the hand, and it will not be affected by the ambient light as easily as ordinary cameras. Moreover, it is cheap, convenient and efficient to use, and is favored by more and more gesture recognition researchers. Scholars at home and abroad have done a lot of research [10–17], and also made many achievements.

With the development of deep learning and computer vision, gesture recognition based on computer vision has begun to rise. The data acquisition of computer vision technology mainly uses ordinary digital cameras. Compared with data gloves and body sensing equipment, it is cheaper, more convenient and more consistent with people's use habits. Scholars all over the world have done a lot of work in this area. In 2015, Molchanov first used two subnets to extract multi-scale gesture spatio-temporal features at the CVPR 2015 meeting, and then sent the gesture spatio-temporal features to the 3D Convolutional Neural Network (3D CNN) for dynamic gesture recognition, with 77.5% accuracy in the VIVA dataset [18].

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In 2016, Miao Qiguang from Xi'an University of Electronic Science and Technology extracted features by inputting two-dimensional pictures and depth videos into 3D CNN, and then classified them through SVM, and achieved an accuracy rate of 56.9% [19]. In 2017, Eibadawy et al. from Eingsams University in Egypt developed a system for recognizing isolated words in Arabic sign language based on 3D CNN, and the recognition accuracy of 25 Arabic words by inputting pictures reached 98% [20]. In 2018, Pu et al. of the University of Science and Technology of China integrated the expansive convolutional network and 3D CNN to recognize continuous gestures, and achieved 83.2% accuracy in 100 Chinese sign languages [21]. In 2018, Bantupalli K from Kenneth State University in the United States used Inception to recognize 100 kinds of American isolated word sign language, and the recognition accuracy reached 93.0% [22]. In 2019, Hernandez V from Tokyo Agricultural University first scaled and smoothed the input data, and then put it into a network combining CNN and LSTM to realize the recognition of 19 American isolated word sign languages, and the recognition accuracy reached 89.3% [23]. In 2020, Li Dan from Chongqing University of Posts and Telecommunications will send gesture samples into a 3D CNN and LSTM connected network with residual structure, extract features and classify them through Softmax. The accuracy of 100 gestures is 95.5% [24]. In 2021, M Saleem proposed a two-dimensional convolutional network model to recognize hand posture. The recognition accuracy of three open data sets Kaggle, First Person and Dexter is 99.99%, 99.48% and 98% respectively [25].

In 2019, Google publicly released MediaPipe, an open source project used to build a machine learning pipeline for processing video, audio and other time series data. This cross platform framework is applicable to desktop/server, Android, iOS and embedded devices, such as Raspberry Pi and Jetson Nano. It supports many common AI functions, such as face detection, FaceMesh, portrait segmentation, gesture tracking, human posture estimation, and hair coloring. Therefore, researchers in various countries have conducted various AI application researches based on MediaPipe. This paper proposes a gesture recognition system based on the MediaPipe open source framework. This system uses the OpenCV library based on MediaPipe to detect 21 key points of the gesture, identify the gesture, and output the letter represented by the gesture by connecting the real-time surveillance camera or other means.

## 2 SYSTEM DESIGN

### 2.1 System flow chart

The system uses Opencv and MediaPipe. First, Opencv and MediaPipe are used to preprocess images and identify key points, and then specific gesture recognition is carried out through gesture recognition module. The recognition accuracy of the system can be improved through rough gesture recognition and fine gesture recognition. The following figure 1 shows the basic flow chart of system design:

### 2.2 Introduction to system operation environment

The Mediapipe used in this system is an open source polyhedron machine learning framework developed by Google Academic Research Institute. MediaPipe is mainly used to process various forms

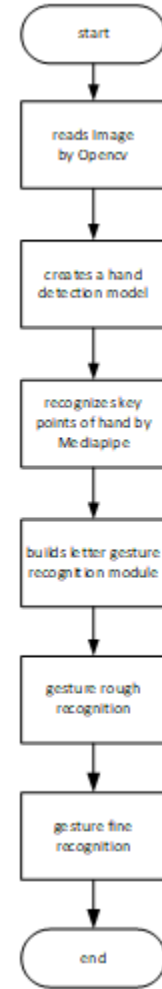


Figure 1: System Flow Chart

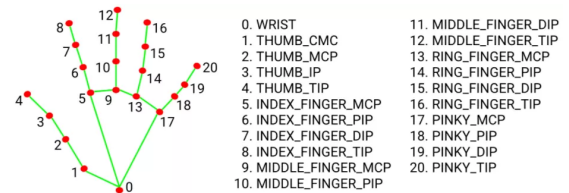


Figure 2: Hand joint point marking and sorting

of data sources (such as video, audio and various other sensor data and time series data). It contains many models and machine learning algorithms, such as pose, face detection, iris and so on.

MediaPipe is highly compatible, can run on multiple platforms, and supports computers to use GPU processing. With MediaPipe, the image to be processed can be decomposed into multiple data flow pipelines represented by graphics modules.

**Table 1: Gesture recognition classification result confusion matrix - 1**

| Predict gestures Real gestures | A   | B   | C   | D   | E   | F   | G   | H   |
|--------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|
| A                              | 257 | 0   | 0   | 0   | 0   | 0   | 1   | 0   |
| B                              | 0   | 280 | 0   | 0   | 0   | 0   | 0   | 0   |
| C                              | 0   | 0   | 284 | 0   | 0   | 0   | 0   | 0   |
| D                              | 2   | 0   | 0   | 266 | 0   | 0   | 0   | 0   |
| E                              | 5   | 0   | 0   | 0   | 277 | 0   | 0   | 0   |
| F                              | 0   | 2   | 0   | 0   | 0   | 281 | 0   | 0   |
| G                              | 0   | 0   | 13  | 0   | 0   | 0   | 264 | 17  |
| H                              | 0   | 4   | 0   | 0   | 0   | 0   | 14  | 278 |
| I                              | 0   | 0   | 0   | 6   | 0   | 0   | 0   | 0   |
| J                              | 0   | 0   | 0   | 6   | 0   | 0   | 1   | 0   |
| K                              | 0   | 0   | 1   | 0   | 0   | 0   | 0   | 0   |
| L                              | 0   | 4   | 0   | 9   | 0   | 2   | 0   | 0   |
| M                              | 9   | 0   | 0   | 0   | 3   | 0   | 0   | 0   |
| N                              | 12  | 0   | 4   | 0   | 0   | 0   | 0   | 4   |
| O                              | 0   | 0   | 9   | 0   | 1   | 0   | 6   | 0   |
| P                              | 0   | 0   | 0   | 0   | 0   | 0   | 4   | 0   |
| Q                              | 0   | 0   | 2   | 0   | 0   | 3   | 0   | 0   |
| R                              | 0   | 0   | 2   | 0   | 0   | 0   | 0   | 3   |
| S                              | 9   | 0   | 0   | 0   | 0   | 0   | 0   | 0   |
| T                              | 0   | 0   | 0   | 0   | 5   | 0   | 1   | 0   |
| U                              | 0   | 0   | 2   | 0   | 0   | 0   | 0   | 0   |
| V                              | 0   | 0   | 0   | 0   | 2   | 0   | 0   | 0   |
| W                              | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 2   |
| X                              | 0   | 0   | 0   | 7   | 0   | 0   | 0   | 0   |
| Y                              | 0   | 0   | 0   | 0   | 6   | 0   | 0   | 0   |
| Z                              | 0   | 0   | 0   | 8   | 0   | 0   | 0   | 0   |

### 2.3 Gesture key point extraction

This system uses Mediapipe Hands as the key point extraction scheme of gesture. It is mainly composed of two neural network models: BlazePalm Detector and Hand Landmark Model. The output of the BlazePalm Detector model is processed and converted to the coordinates of the detection box containing only the palm, and the key point positions of the palm are output. Using the palm detection box, middle finger joint and wrist joint outputted from the model, Mediapipe Hand rotates and magnifies the detection box to obtain a full hand detection box. The Hand Landmark Model converts the images in the reduced full hand detection frame into 3D coordinates, left and right chirality and gesture scores of 21 key points of the whole hand, as shown in Figure 2. In real-time applications, once the gesture is detected by the palm detection model, the subsequent full hand detection frame is generated by the gesture key point model.[26]

With the help of MediaPipe, the data stream can be constructed into a modular group diagram, and the video or audio stream data can be input into the graph. The graph model pipes constructed by various functional modules can be used to process these data and output the final result data from the graph. Make the overall processing flow more smooth. Therefore, developers can use MediaPipe as an environment for iteratively optimizing and improving their applications, and the final results can be reproduced on different devices and platforms.

## 3 SYSTEM IMPLEMENTATION

This system uses the letter recognition module to solve the problem of actual letter recognition in a step-by-step way, that is, first roughly classify the recognized hand shape, and then identify the specific meaning of the gesture through the specific shape of the finger. This step-by-step processing method can effectively reduce the time complexity of the program and improve the running efficiency of the program.

### 3.1 Rough classification of gesture recognition

This system will roughly classify gestures according to the position relationship between the fingertip of each finger and each joint, and store the posture of each finger for later refinement and classification. The basic gestures of gestures are classified by the basic hand type of gestures. After rough classification of gestures, the space complexity and time complexity required for subsequent subdivision judgment can be reduced. Algorithm explanation: identify key points of gesture through Mediapipe Hands. Compare the coordinates of tip TIP and tip TIP joint MCP with finger joint PIP, and store the comparison results. The code is as follows:

### 3.2 Fine classification of gesture recognition

This system describes the refined classification algorithm through some code examples. Through rough classification, we can get the general state of a full clenched fist, which is the general state of A, M, N, S gestures. To distinguish these four gestures, it is necessary

**Table 2: Gesture recognition classification result confusion matrix - 2**

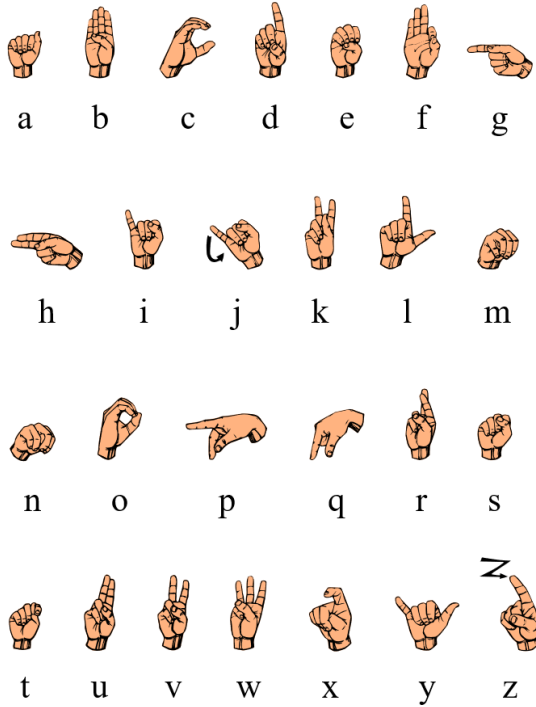
| Predict gestures Real gestures | I   | J   | K   | L   | M   | N   | O   | P   |
|--------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|
| A                              | 0   | 0   | 0   | 0   | 12  | 11  | 0   | 0   |
| B                              | 0   | 0   | 8   | 0   | 0   | 0   | 0   | 0   |
| C                              | 3   | 0   | 0   | 0   | 0   | 0   | 13  | 0   |
| D                              | 0   | 0   | 2   | 9   | 3   | 0   | 0   | 4   |
| E                              | 0   | 0   | 0   | 6   | 0   | 0   | 0   | 0   |
| F                              | 0   | 0   | 3   | 0   | 1   | 0   | 0   | 3   |
| G                              | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 4   |
| H                              | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 2   |
| I                              | 257 | 8   | 0   | 0   | 0   | 8   | 0   | 5   |
| J                              | 12  | 271 | 0   | 0   | 0   | 1   | 0   | 0   |
| K                              | 0   | 0   | 280 | 0   | 0   | 0   | 0   | 0   |
| L                              | 0   | 0   | 0   | 262 | 0   | 0   | 0   | 0   |
| M                              | 0   | 0   | 0   | 0   | 256 | 14  | 0   | 0   |
| N                              | 5   | 3   | 0   | 0   | 12  | 250 | 0   | 0   |
| O                              | 0   | 0   | 0   | 0   | 0   | 0   | 267 | 0   |
| P                              | 3   | 0   | 0   | 3   | 0   | 0   | 0   | 288 |
| Q                              | 0   | 0   | 7   | 0   | 2   | 0   | 0   | 0   |
| R                              | 0   | 0   | 0   | 0   | 0   | 1   | 0   | 0   |
| S                              | 7   | 0   | 0   | 0   | 9   | 8   | 3   | 0   |
| T                              | 3   | 0   | 4   | 0   | 6   | 0   | 0   | 0   |
| U                              | 0   | 0   | 8   | 0   | 3   | 0   | 0   | 0   |
| V                              | 4   | 0   | 7   | 0   | 0   | 4   | 0   | 3   |
| W                              | 0   | 0   | 0   | 8   | 0   | 0   | 0   | 0   |
| X                              | 3   | 0   | 0   | 9   | 5   | 0   | 0   | 0   |
| Y                              | 16  | 0   | 2   | 0   | 5   | 0   | 0   | 0   |
| Z                              | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   |

**Table 3: Gesture recognition classification result confusion matrix - 3**

| Predict gestures Real gestures | Q   | R   | S   | T   | U   | V   | W   | X   | Y   | Z   |
|--------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| A                              | 0   | 0   | 19  | 0   | 0   | 0   | 0   | 0   | 0   | 0   |
| B                              | 0   | 0   | 12  | 0   | 0   | 0   | 0   | 0   | 0   | 0   |
| C                              | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   |
| D                              | 0   | 0   | 0   | 0   | 0   | 6   | 0   | 8   | 0   | 0   |
| E                              | 0   | 0   | 0   | 12  | 0   | 0   | 0   | 0   | 0   | 0   |
| F                              | 0   | 0   | 1   | 2   | 0   | 3   | 0   | 4   | 0   | 0   |
| G                              | 0   | 0   | 0   | 0   | 2   | 0   | 0   | 0   | 0   | 0   |
| H                              | 0   | 0   | 0   | 0   | 0   | 0   | 2   | 0   | 0   | 0   |
| I                              | 0   | 0   | 0   | 1   | 0   | 0   | 0   | 0   | 15  | 0   |
| J                              | 0   | 4   | 0   | 0   | 0   | 0   | 2   | 0   | 0   | 3   |
| K                              | 0   | 4   | 0   | 0   | 7   | 8   | 0   | 0   | 0   | 0   |
| L                              | 0   | 0   | 0   | 0   | 0   | 5   | 0   | 11  | 0   | 7   |
| M                              | 0   | 0   | 16  | 0   | 2   | 0   | 0   | 0   | 0   | 0   |
| N                              | 0   | 0   | 10  | 0   | 0   | 0   | 0   | 0   | 0   | 0   |
| O                              | 0   | 0   | 0   | 0   | 0   | 12  | 5   | 0   | 0   | 0   |
| P                              | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 2   | 0   | 0   |
| Q                              | 276 | 0   | 0   | 3   | 0   | 5   | 0   | 2   | 0   | 0   |
| R                              | 0   | 275 | 0   | 0   | 19  | 0   | 0   | 0   | 0   | 0   |
| S                              | 0   | 0   | 253 | 6   | 0   | 0   | 0   | 0   | 5   | 0   |
| T                              | 0   | 8   | 4   | 269 | 0   | 0   | 0   | 0   | 0   | 0   |
| U                              | 0   | 9   | 0   | 0   | 265 | 13  | 0   | 0   | 0   | 0   |
| V                              | 0   | 8   | 0   | 0   | 9   | 263 | 0   | 0   | 0   | 0   |
| W                              | 0   | 0   | 7   | 0   | 0   | 0   | 283 | 0   | 0   | 0   |
| X                              | 0   | 6   | 0   | 0   | 0   | 0   | 0   | 270 | 0   | 0   |
| Y                              | 0   | 0   | 0   | 4   | 0   | 0   | 0   | 0   | 267 | 0   |
| Z                              | 0   | 4   | 0   | 0   | 3   | 8   | 0   | 0   | 0   | 277 |

**Table 4: Precision and Recall of each gesture and F1 measures1**

|   | Precision | Recall | F1-Measure |
|---|-----------|--------|------------|
| A | 87.41%    | 85.67% | 86.53%     |
| B | 96.55%    | 93.33% | 94.92%     |
| C | 89.59%    | 94.67% | 92.06%     |
| D | 88.08%    | 88.67% | 88.37%     |
| E | 94.22%    | 92.33% | 93.27%     |
| F | 98.25%    | 93.67% | 95.90%     |
| G | 90.72%    | 88.00% | 89.34%     |
| H | 91.45%    | 92.67% | 92.05%     |
| I | 82.11%    | 85.67% | 83.85%     |
| J | 96.10%    | 90.33% | 93.13%     |
| K | 87.23%    | 93.33% | 90.18%     |
| L | 88.22%    | 87.33% | 87.77%     |
| M | 81.53%    | 85.33% | 83.39%     |
| N | 84.18%    | 83.33% | 83.75%     |
| O | 94.35%    | 89.00% | 91.60%     |
| P | 93.20%    | 96.00% | 94.58%     |
| Q | 100.00%   | 92.00% | 95.83%     |
| R | 86.48%    | 91.67% | 89.00%     |
| S | 78.57%    | 84.33% | 81.35%     |
| T | 90.57%    | 89.67% | 90.12%     |
| U | 86.32%    | 88.33% | 87.31%     |
| V | 81.42%    | 87.67% | 84.43%     |
| W | 96.92%    | 94.33% | 95.61%     |
| X | 90.91%    | 90.00% | 90.45%     |
| Y | 93.03%    | 89.00% | 90.97%     |
| Z | 96.52%    | 92.33% | 94.38%     |

**Figure 3: American ASL Gesture****Algorithm 1** CODE 1: rough classification

```

import cv2
import mediapipe as mp
mp_hands = mp.solutions.hands
hands = mp_hands.Hands()
mp_draw = mp.solutions.drawing_utils
cap = cv2.VideoCapture(1)
finger_tips = [8, 12, 16, 20]
index_tip = 8
middle_tip = 12
ring_tip = 16
pinky_tip = 20
thumb_tip = 4
while True:
    ret, img = cap.read()
    img = cv2.flip(img, 1)
    h, w, c = img.shape
    results = hands.process(img)
    if results.multi_hand_landmarks:
        for hand_landmark in results.multi_hand_landmarks:
            lm_list = []
            for id, lm in enumerate(hand_landmark.landmark):
                lm_list.append(lm)
                finger_fold_status = []
                finger_fold_GH = []
                finger_above_palm = []
                thumb_between = []
                for tip in finger_tips:
                    x, y = int(lm_list[tip].x*w), int(lm_list[tip].y*h)
                    if lm_list[tip].y > lm_list[tip-3].y:
                        finger_fold_status.append(True)
                else:
                    finger_fold_status.append(False)
                if lm_list[tip].y > lm_list[tip-2].y:
                    finger_above_palm.append(True)
                else:
                    finger_above_palm.append(False)
                if lm_list[tip].x > lm_list[tip-3].x:
                    finger_fold_GH.append(True)
                else:
                    finger_fold_GH.append(False)
                if lm_list[tip].y <= lm_list[tip-3].y:
                    thumb_between.append(True)
                else:
                    thumb_between.append(False)

```

to make a key judgment on the thumb. The specific meaning of the gesture can be obtained by analyzing and judging the position of the thumb and each joint on the thumb.

Refine classification code (partial)

As shown in Figure 3, American Sign Language (ASL) gesture recognition gesture is used in program design.

**4 EXPERIMENTAL CERTIFICATION**

In order to test the accuracy of gesture recognition, several subjects were used to express various sign language gestures, and real-time detection and judgment of gestures were carried out. The verification test was conducted according to the 26 ASL gestures designed. The real-time detection image is saved, and the data set is expanded through image enhancement, translation, rotation and noise adding.

**Algorithm 2** CODE 2: refined classification

---

```

if all(finger_fold_status):
if lm_list[thumb_tip].y < lm_list[thumb_tip - 1].y < lm_
list[thumb_tip - 2].y and lm_list[thumb_tip].x < lm_
list[thumb_tip - 1].x <= lm_list[thumb_tip - 2].x:
cv2.putText(img, 'A', (20, 30), cv2.FONT_HERSHEY_SIMPLEX, 1,
(0, 255, 0), 3)
if lm_list[thumb_tip].x <= lm_list[ring_tip].x and lm_
list[thumb_tip].x > lm_list[thumb_tip - 1].x:
cv2.putText(img, 'S', (20, 30), cv2.FONT_HERSHEY_SIMPLEX, 1,
(0, 255, 0), 3)
if lm_list[thumb_tip].x > lm_list[pinky_tip].x:
cv2.putText(img, 'M', (20, 30), cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 3)
if lm_list[thumb_tip].x < lm_list[pinky_tip].x and lm_
list[thumb_tip].x > lm_list[ring_tip].x:
cv2.putText(img, 'N', (20, 30), cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 3)

```

---

And the same gesture is tested in different environments and different testers. Finally, the accuracy confusion matrix of gesture recognition is obtained. As shown in Table 1. In the table, the accuracy, recall and F1 measures are used to evaluate the performance of gesture recognition algorithm

The accuracy, recall and F1 measures are used to evaluate the performance of gesture recognition algorithm, and Table 4 is obtained, where:

$$\text{Precision} : P = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} : R = \frac{TP}{TP + FN} \quad (2)$$

Equation 1,2:

TP (True Positive): the true case, that is, the part that is predicted to be positive and true to be positive;

FP (False Positive): false positive cases, that is, the predicted positive part and the true negative part;

FN (False Negative): false negative cases, that is, the prediction is negative, and the true is positive.

$$\text{F1measure} : F1 = \frac{2 * P * R}{P + R} = \frac{2 * TP}{TP + FP + TP + FN} \quad (3)$$

Table 4 shows that the recognition accuracy of the system is above 81%, and the recall rate is above 83%, which can be applied to gesture recognition related applications of English letters.

## 5 CONCLUSION

This paper proposes an application system based on MediaPipe for intelligent recognition of 26 letter gestures in English. The system uses the images captured by a monocular camera, reads the images through OpenCV, recognizes 21 key points of the hand based on MediaPipe, constructs a gesture recognition module for English letters, and recognizes English letters through rough gesture recognition algorithm and fine gesture recognition algorithm. The experimental results show that the recognition accuracy and recall rate of the system are above 81%, which can be applied to gesture recognition related applications of English letters. This system also

has some limitations. It cannot recognize the English words of gestures, which limits the application of gesture recognition. Gesture recognition is widely used. With the development of image recognition and artificial intelligence technology, it is expected that more application results will appear in the future.

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